THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The line $17x + 18y = 306$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 155 **(b)** 153 **(c)** 152 **(d)** 154

2. A taxi fare is ₹243 for 5 km and ₹425 for 12 km (fare is a linear function of distance). The fare for 38 km (in ₹) is: 1

(a) 1127 **(b)** 1214 **(c)** 1101 **(d)** 1075

3. The line $4x + y = 9$ is parallel to the line $kx + 5y = 11$. The value of k is: 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 19

4. The points $(-5, 19)$, $(-4, 18)$ and $(-3, k)$ are collinear. The value of k is: 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 16

5. The line $kx + 2y = 16$ passes through $(1, 4)$. Its slope is: 1

(a) -4 **(b)** -2 **(c)** -3 **(d)** -5

6. The line $9x + 5y = 45$ has intercepts p on the x-axis and q on the y-axis. The value of $p + q$ is: 1

(a) 14 **(b)** 15 **(c)** 59 **(d)** 13

7. Using $F = \frac{9}{5}C + 32$, the temperature 90°C in degrees Fahrenheit is: 1

(a) 194 **(b)** 195 **(c)** 193 **(d)** 203

8. The slope of the line through $(12, 22)$ and $(-24, 23)$ is: 1

(a) $\frac{71}{36}$ **(b)** $-\frac{1}{36}$ **(c)** $\frac{35}{36}$ **(d)** $-\frac{37}{36}$

9. **Assertion (A):** The pair $(2, 3)$ is a solution of $3x + 2y = 12$. 1

Reason (R): A pair (x, y) is a solution of an equation if substituting it makes both sides equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A linear equation in two variables has infinitely many solutions. 1

Reason (R): The graph of a linear equation in two variables is a straight line.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The line $kx + 3y = 27$ passes through the point $(1, 6)$. Find k, the slope and the y-intercept of the line. 2

12. Find the area of the triangle formed by the line $9x + 15y = 135$ and the coordinate axes. 2

13. Find the equation of the line through $(-12, -24)$ and $(-9, -21)$ in the form $y = mx + c$. 2

14. The line $kx + 4y = 36$ passes through the point $(3, 6)$. Find k, the slope and the y-intercept of the line. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A line cuts the x-axis at $(18, 0)$ and the y-axis at $(0, 17)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

16. A water supplier charges a fixed amount plus a rate per kilolitre. For 4 kL the bill is ₹107 and for 11 kL it is ₹212. Find the equation of the bill, and the usage whose bill is ₹317. Also find the bill for no usage. 3

17. Find k if the points $(19, -9)$, $(21, -13)$ and $(23, k)$ lie on one straight line. 3

OR

A line cuts the x-axis at $(15, 0)$ and the y-axis at $(0, 18)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

18. A water supplier charges a fixed amount plus a rate per kilolitre. For 4 kL the bill is ₹116 and for 10 kL it is ₹212. Find the equation of the bill, and the usage whose bill is ₹324. Also find the bill for no usage. 3

OR

Find k if the points $(-18, -17)$, $(-17, -15)$ and $(-16, k)$ lie on one straight line. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rahul has Rs 100 to spend at a stationery shop in Delhi. A pen costs Rs 5 and a notebook costs Rs 20. He buys x pens and y notebooks and spends exactly Rs 100. 5

(i) Write the linear equation in x and y.

(ii) Is $(4, 4)$ a solution?

(iii) If he buys 2 notebooks, how many pens does he buy?

(iv) Find the x-intercept and y-intercept of the line and interpret them.

20. A line passes through $(0, 2)$ and $(1, 6)$. Find its equation, then find k if the point $(k, 30)$ lies on it, and the x-intercept. 5

OR

A taxi fare is ₹222 for 4 km and ₹310 for 8 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 38 km. 5

END OF QUESTION PAPER — Total: 40 Marks